

environment, the N-Gage Arena. This online service is an environment where N-Gage owners can compete, communicate and obtain special benefits for their activities. In traditional game console markets similar online services have become very popular (e.g. X-Box Live). Mobile online game services are expected to follow this trend.

An interesting dimension to mobile multiplayer gaming is brought by exploiting the location-identification features of the mobile networks. A Swedish company called 'It's Alive' has created a multiplayer game called Botfighters. In it players chase each other to various cellular network locations. The experiences from Botfighters indicate that there are interesting opportunities for combining real-life action and a virtual game play. German Jamba has created another interesting new mobile game concept. In their game "Attack of the Killer Virus" player shoots viruses/monsters projected to a real-life environment shown through a lens of a camera phone. A player has to move around with the camera to destroy the viruses. Similar kinds of combinations of cellular network technology and real-life experience will with little doubt increase in the future.

An innovative dimension to multiplayer gaming is created by a combination of television broadcasting and mobile phones. The players compete against each other by SMS commands in various kind of games, e.g. in shooting, skills, sports etc.. The users are charged by move, i.e. by SMS. These games have become surprisingly popular throughout Northern Europe-most probably due to the famous Warhol's 15 minutes (in mobile 15 seconds) of fame phenomena. People love to see their messages and game actions to be displayed to the large audiences which game-shows on television may gather. In the near future, similar kind of gaming communities may also appear around digital television applications in terrestrial, satellite and cable television networks.

Mobile Games E-Content Report 3: An integrating report by ACTeN (Anticipating Content Technology Need)